16. The Cost of Lumber, continued The analysis of variance F-test in Exercise 15 (Section 11.2) indicated that there was a difference in the average cost of lumber for the four states. The following information from that exercise is given in the table:

Sample Means	$\bar{x}_1 = 262.2$	MSE	41.25
	$\bar{x}_2 = 234.8$	Error *df*:	16
	$\bar{x}_3 = 251.6$	n_i:	5
	$\bar{x}_4 = 268.6$	k:	4

Use Tukey's method for paired comparisons to determine which means differ significantly from the others at the $\alpha = .01$ level.

Q16

$q_{4,16;0.01} \approx 5.19$

$W = q_{4,16} \cdot \sqrt{\frac{MSE}{n_i}}$

$W = 5.19\sqrt{\frac{41.25}{5}} = 14.91$

$|\bar{x}_4 - \bar{x}_2| = |268.6 - 234.81| = 33.8 > 14.91$ Yes

$|\bar{x}_4 - \bar{x}_3| = |268.6 - 251.61| = 17.0 > 14.91$ Yes

$|\bar{x}_1 - \bar{x}_2| = |262.2 - 234.8| = 27.4 > 14.91$ Yes

$|\bar{x}_1 - \bar{x}_3| = |262.2 - 251.61| = 10.6 < 14.91$ No

$|\bar{x}_3 - \bar{x}_2| = |251.6 - 234.81| = 16.8 > 14.91$ Yes

$|\bar{x}_4 - \bar{x}_1| = |268.6 - 262.2| = 6.4 < 14.91$ NO

At $\alpha = 0.01$, (4 and 2), (4 and 3), (1 and 2) and (3 and 2) are diff. Since 2 appears in all the test, it is significantly different.

DATA SET DS1113

17. GRE Scores The quantitative reasoning scores on the Graduate Record Examination (GRE)[3] were recorded for students admitted to three different graduate programs at a local university.

x_1		Graduate Program x_2		x_3	
Life Sciences		Physical Sciences		Social Sciences	
630	660	660	760	440	540
640	660	640	670	330	450
470	480	720	700	670	570
600	650	690	710	570	530
580	710	530	450	590	630

a. Do these data provide sufficient evidence to indicate a difference in the mean GRE scores for applicants admitted to the three programs?

b. Find a 95% confidence interval for the difference in mean GRE scores for Life Sciences and Physical Sciences.

c. If you find a significant difference in the average GRE scores for the three programs, use Tukey's method for paired comparisons to determine which means differ significantly from the others. Use $\alpha = .05$.

Q17 a)

$\bar{x}_1 = 608$ $\bar{x}_2 = 653$ $\bar{x}_3 = 532$ $\bar{G} = 597.67$

$n_1 = 10$ $n_2 = 10$ $n_3 = 10$

$TSS = \sum (x_{ij} - \bar{G})^2 = 300736.67$

$SST = \sum_{i=1}^{a} (\bar{T}_i - \bar{G})^2 = 74806.67$

$SSE = TSS - SST = 225930$

$MST = \frac{74806.67}{2} = 37403.335$

$MSE = \frac{225930}{27} = 8367.78$

$F = \frac{MST}{MSE} = \frac{37403.335}{8367.78} = 4.47$

$H_0: \mu_1 = \mu_2 = \mu_3$ H_1: At least one is diff

$\alpha = 0.05$

R.R. $F_{2,27;0.05} \approx 3.35$

$4.47 > 3.35$

$\therefore$ Rej H_0

b) $(\bar{x}_1 - \bar{x}_2) \pm q_{3,27;0.05}\sqrt{\frac{MSE}{10}} = (-146.42, 56.42)$

c) $W = q_{3,27}\sqrt{\frac{MSE}{n}} = q_{3,27,0.05} \cdot \sqrt{\frac{8367.78}{10}} = 101.42$

$|\bar{x}_1 - \bar{x}_2| = |608 - 653| = 45 < 101.42$ No

$|\bar{x}_1 - \bar{x}_3| = |608 - 532| = 76 < 101.42$ No

$|\bar{x}_2 - \bar{x}_3| = |653 - 532| = 121 > 101.42$ Yes

$\therefore$ Only Physical sciences and Social sciences have significantly diff means.

DATA SET DS1121

24. Where to Shop? Do you shop at the grocery store closest to home or do you look for the store that has the best prices? We compared the regular prices at four different grocery stores for eight items purchased on the same day.

	Stores			
Items	Vons	Ralphs	Stater Bros	WinCo
Salad mix, 360 g bag	3.99	2.79	1.99	1.78
Hillshire Farm® Beef Smoked Sausage, 420 g	4.29	4.29	3.99	2.50
Kellogg's Raisin Bran®, 750 g	4.49	5.49	4.49	3.15
Kraft® Philadelphia® Cream Cheese, 250 g	2.99	3.19	2.79	1.48
Kraft® Ranch Dressing, 450 mL	3.19	3.49	3.49	1.48
Treetop® Apple Juice, 2 L	2.99	3.49	3.49	1.58
Dial® Bar Soap, Gold, 125 g	5.99	6.49	5.79	5.14
Jif® Peanut Butter, Creamy, 850 g	5.15	5.49	4.79	4.34

a. What are the blocks and treatments in this experiment?

b. Do the data provide evidence to indicate that there are significant differences in prices from store to store? Support your answer statistically using the ANOVA printout that follows.

c. Are there significant differences from block to block? Was blocking effective?

ANOVA: Price versus Item, Store

Analysis of Variance for Price

Source	DF	SS	MS	F	P
Store	3	13.192	4.3974	26.52	0.000
Item	7	40.995	5.8564	35.32	0.000
Error	21	3.482	0.1658		
Total	31	57.669			

Model Summary

S	R-sq	R-sq(adj)
0.407181	93.96%	91.09%

Q24 a) Blocks: items $b=8$ $N=32$ Treatments: stores $k=4$

b) $H_0: \mu_1=\mu_2=\mu_3=\mu_4$ H_1: not same

$\alpha=0.05$

$F^* = 26.52$

R.R. $F_{3,21;0.05} \approx 3.07$

$26.52 > 3.07$

$\therefore$ Reject H_0. There is sign. diff.

c) $H_0: \mu_1=\mu_2=\mu_3=\mu_4=\mu_5=\mu_6=\mu_7=\mu_8$

H_1: not same

$\alpha=0.05$

$F^* = 26.52$

R.R. $F_{7,21;0.05} \approx 2.49$

$26.52 > 2.49$

$\therefore$ Reject H_0, there is sign. diff.

25. Where to Shop?, continued Refer to Exercise 24. The printout that follows provides the average costs of the selected items for the $k = 4$ stores.

Means

Store	N	Price
Ralphs	8	4.34000
Stater Bros	8	3.85250
Vons	8	4.13500
WinCo	8	2.68125

a. What is the appropriate value of $q_{.05}(k, df)$ for testing for differences among stores?

b. What is the value of $\omega = q_{.05}(k, df)\sqrt{\frac{MSE}{b}}$?

c. Use Tukey's pairwise comparison test among stores used to determine which stores differ significantly in average prices of the selected items.

Q25 a) $q_{4,21;0.05}$

b) $W = q_{4,21;0.05}\sqrt{\frac{MSE}{N}} = 3.98 \cdot \sqrt{\frac{0.1658}{8}} = 0.5688$

c) $|\bar{x}_1-\bar{x}_2| = |4.34-3.85250| = 0.4875 < W$ No

$|\bar{x}_1-\bar{x}_3| = |4.34-4.135| = 0.205 < W$ No

$|\bar{x}_1-\bar{x}_4| = |4.34-2.68125| = 1.658 > W$ Yes

$|\bar{x}_2-\bar{x}_3| = |3.85250-4.135| = 0.28 < W$ No

$|\bar{x}_2-\bar{x}_4| = |3.85250-2.68125| = 1.172 > W$ Yes

$|\bar{x}_3-\bar{x}_4| = |4.135-2.68125| = 1.453 > W$ Yes

Since $\bar{x}_4$ appears in all, WinCo is sign. diff. than others

DATA SET DS1124

26. Rocket Propellants An experiment was conducted to compare three mixtures of components in making rocket propellant. Five samples of each mixture were obtained for testing. Five investigators measured the propellant thrust in pounds for each of the mixtures, with the mixtures presented in random order.

	Mixtures		
Investigators	1	2	3
1	42	39	45
2	37	36	40
3	43	48	50
4	41	47	52
5	39	42	47

a. Identify the treatments and the blocks in this experiment and provide an appropriate analysis of variance.

b. Is there sufficient evidence to conclude there is a significant difference in the propellant thrust for the mixtures? Use $\alpha = .01$.

c. Was blocking effective? Use the p-value to draw your conclusions and justify your answer.

d. Use Tukey's pairwise comparisons with $\alpha = .01$ to determine which propellant mixtures differed from others. Summarize your results.

Q26 a) Blocks: investigator $b=5$ Treatment: mixture $k=3$

b) $N=15$ $N_1=5$ $N_2=5$ $N_3=5$

$T_1=202$ $T_2=232$ $T_3=234$

$\bar{x}_1=40.4$ $\bar{x}_2=46.4$ $\bar{x}_3=46.8$

$\bar{x}_G=43.2$

$TSS = \Sigma(x_{ij}-\bar{x}_G)^2 = 322.4$

$SST = \sum_{i=1}^{k} n_i(\bar{x}_i-\bar{x}_G)^2 = 107.2$

$SSB = 176.4$

$SSE = 322.4 - 107.2 - 176.4 = 38.8$

Source	SS	df	MS	F
Treatment	107.2	2	MST = 107.2/2 = 53.6	MST/MSE = 11.05
Block	176.4	4	MSB = 176.4/4 = 44.1	MSB/MSE = 9.09
Error	38.8	8	MSE = 38.8/8 = 4.85	
Total	322.4	14		

$H_0: \mu_1=\mu_2=\mu_3$ ... H_1: not same

$\alpha=0.05$

$F^* = 11.05$

R.R. $F_{2,8;0.01} = 8.65$

$\therefore$ $11.05 > 8.65$, there is sign. diff.

c) $P(F_{4,8} > 9.09) = 0.0045 < 0.01$

$\therefore$ There is diff.

d) $W = q_{4,8;0.01} \cdot \sqrt{\frac{MSE}{n}} = 5.55$

$|\bar{x}_1-\bar{x}_2| = |40.4-46.4| = 2 < 5.55$ No

$|\bar{x}_1-\bar{x}_3| = |40.4-46.8| = 6.4 > 5.5$ Yes

$|\bar{x}_2-\bar{x}_3| = |46.4-46.8| = 0.4 < 5.5$ No

$\therefore$ mixture 1 and 3 has significant diff.